

Finding strongly lensed gravitational-wave pairs in ground-based detector catalogs

Fan Zhang^{1,2}, Qikai Zhang¹, Qiyuan Yang³, Jiaqing Huang⁴, Yong Yuan⁵ & Xilong Fan^{3,*}

¹State Key Laboratory of Ocean Sensing and Ocean College, Zhejiang University, Zhoushan 316021, China

²Kavli Institute for Astrophysics and Space Research, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

³School of Physics and Technology, Wuhan University, Wuhan 430072, China

⁴Jiangsu Sungo Intelligent Technology Research Center Co., Ltd., Jiangsu 226019, China

⁵Center for Gravitational Wave Experiment, Institute of Mechanics, Chinese Academy of Sciences, Beijing, China

*Corresponding author: xilong.fan@whu.edu.cn

Abstract

Strong gravitational lensing can make one compact-binary merger appear as multiple gravitational-wave events. In third-generation surveys, event pairs grow as N^2 , making lens searches a catalog-level combinatorics problem. We address this problem by embedding multi-detector waveforms with an InceptionTime encoder and ranking companions with time-delay and sky-localization information. We train on simulated singular-isothermal-sphere and point-mass lenses in an Einstein Telescope configuration and test on a 9,000-event catalog. The waveform embedding score retrieves 62.5% at rank one and 85.4% within the top ten. The catalog ranking score raises rank-one retrieval to 97.6% and top-ten recall to 99.9%. Run-matched real-noise deployments on GWTC-3 and the GWTC-4.0 O4a release produce follow-up shortlists on public strain. Stress tests show that pair recovery remains false-association limited at realistic lens rarity. The method provides a scalable refinement layer for Bayesian lens confirmation.

1 Introduction

The direct detection of gravitational waves from compact-binary coalescences opened a new observational window on the Universe[1–4]. Strong lensing is a predicted but unconfirmed signal in this band[5–7]. A massive galaxy or cluster can split one coalescence into multiple images that arrive as separate gravitational-wave events. These images share the same intrinsic source but differ in amplitude, arrival time and phase [8–12]. Finding such pairs would enable precision cosmology, tests of wave propagation and new probes of dark matter[13–17]. No confident detection has yet been reported[18–21].

The scientific goal of this work is to find paired lensed images in a catalog of ground-based gravitational-wave events. Current searches usually begin with pair verification. Given two events, they test whether the posterior parameters, a joint Bayes factor, phase consistency or a machine-learning filter supports a common source[22–29]. Pair verification is well suited to small catalogs, but it leaves a separate discovery problem for the third-generation era. The Einstein Telescope and Cosmic Explorer may detect 10^4 – 10^6 coalescences per year[30–34]. At that scale, every event must be compared with a large catalog, unrelated pairs grow as N^2 , and true lensed systems are expected to be rare[35–37].

We separate the search into two readouts. Companion-image retrieval ranks the catalog for each lensed event and measures where the partner image appears. Complete-pair recovery then asks which high-scoring event pairs remain plausible enough for follow-up. Because the second readout is a rare-pair problem, precision–recall and false-candidate counts are more informative than a ROC curve dominated by true negatives[38–40].

Our implementation follows this division (Fig. 1). An InceptionTime waveform encoder first maps each event to an embedding and gives a fast waveform embedding score. We refer to the downstream physical ranking as the lens-candidate refinement stage. This stage builds pairwise time-delay and sky-localization scores, then combines them with the waveform embedding score to produce the catalog ranking score for candidate pairs. We train and test this design on simulated singular-isothermal-sphere and point-mass lenses in an Einstein Telescope three-interferometer configuration. LIGO Hanford–Livingston controls and HNSW scaling tests are reported in the Supplementary Information. We also test catalog density, lens rarity and GWTC event backgrounds. The pipeline returns a ranked candidate set for Bayesian confirmation.

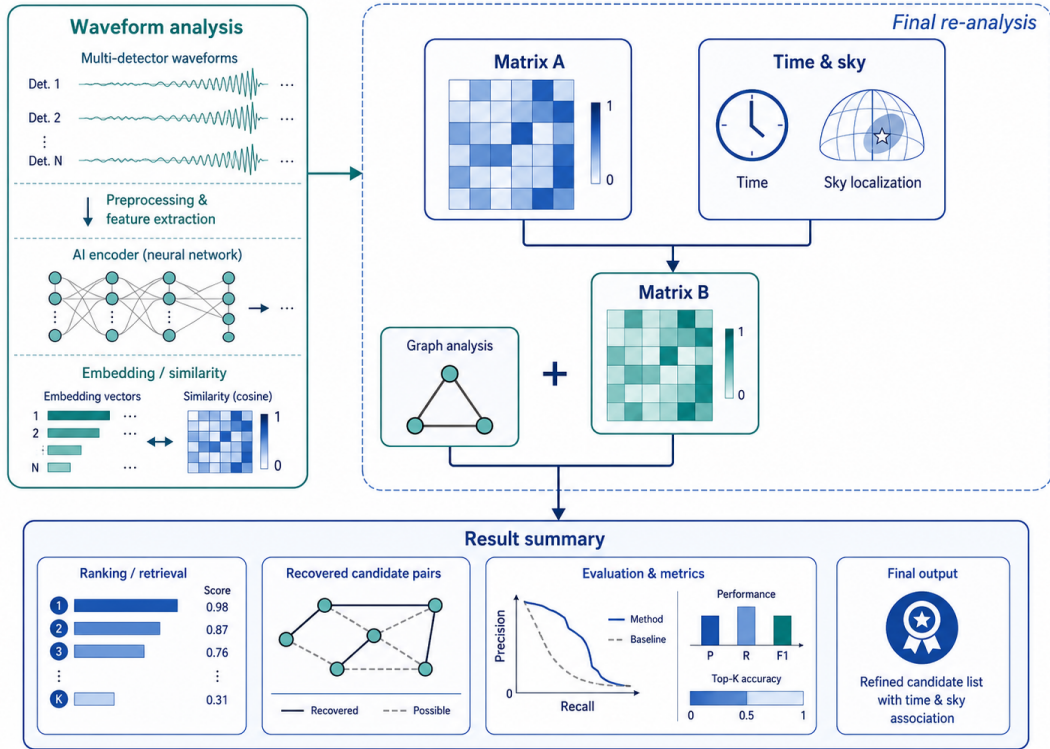


Figure 1. Workflow for catalog retrieval of lensed gravitational-wave pairs. Simulated lensed and isolated events are projected into detector networks, embedded by a waveform encoder, ranked with time-delay and sky-localization information, and assembled into candidate systems. The performance box refers to the balanced ET-3 test catalog; the density and rarity analyses below test the harder astrophysical regime.

Results

The analysis follows the three-part structure set out above. The search target is a pair of lensed gravitational-wave events in a full catalog. The technical implementation first learns a waveform embedding with an InceptionTime encoder, then builds pairwise waveform embedding, time-delay and sky-localization scores for lens-candidate refinement. The main training and test data are simulated SIS, PM and isolated compact-binary events in the ET-3 detector configuration (Methods). We report two related outcomes. Companion-image retrieval asks whether one lensed event retrieves its partner image; this is shown as recall versus rank cutoff. Complete-pair recovery asks whether an accepted event pair remains plausible after thresholding; for this rarer pair-level task we emphasize precision-recall behavior and false-candidate counts.

Companion-image retrieval in the ET-3 catalog

We first evaluated companion-image retrieval in the simulated ET-3 catalog. The catalog contains 9,000 events, made of 3,000 two-image lensed systems (6,000 lensed events) and 3,000 isolated events (Methods). For each lensed event, its companion image is the only correct match. A query is counted as successful at rank k if that companion appears among the top k candidates. Thus the catalog contains 3,000 true unordered

companion pairs among 40,495,500 possible event pairs, whereas recall is averaged over 6,000 directed lensed-event queries, each ranked against 8,999 non-self candidates. All SIS and PM recalls below use this same mixed catalog ranking: the query event is grouped by lens family, but the candidate list still contains the other lens family and the isolated events.

With the waveform embedding score alone, 62.5% of lensed queries rank the companion first and 85.4% place it within the top ten. This is the waveform encoder's role: it supplies a morphology-based candidate ordering, but does not by itself resolve all nearest-neighbour ambiguities. The catalog ranking score combines waveform embedding, time-delay and sky-localization scores and gives the headline mixed catalog result, $R@1 = 0.976$ and $R@10 = 0.999$, over all 6,000 lensed-event queries. The same ranking gives $R@1 = 0.965$ for SIS images and $R@1 = 0.986$ for PM images, with top-ten recall close to unity for both families. Removing the waveform embedding score lowers the overall $R@1$ to 0.715, so the encoder is an essential component rather than an optional addition. This weighted form of the catalog ranking score was more stable than logistic regression, histogram gradient boosting and a small tabular neural network, and is used as the default ranking score in the main analysis.

The family split explains why the auxiliary observables matter. SIS delays are set by the lens velocity dispersion and source-lens alignment, and overlap strongly

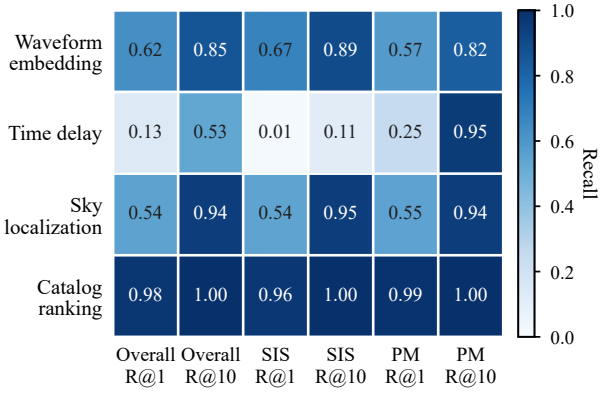


Figure 2. ET-3 companion retrieval by lens family and retrieval score. Recall at rank 1 and rank 10 are shown for the waveform embedding, time-delay, sky-localization and catalog ranking scores. All recalls are computed in the same 9,000-event mixed ET-3 catalog. Overall columns average over all lensed-event queries, whereas SIS and PM columns group the same mixed catalog ranking by the query event’s lens family. The catalog ranking score combines waveform embedding, time-delay and sky-localization scores.

with unrelated event separations in the catalog window. The time-delay score alone therefore gives SIS $R@1 = 0.012$ and $R@10 = 0.111$. PM delays are more distinctive in the adopted mass range, so the time-delay score alone moves most PM companions into the top ten ($R@10 = 0.951$). It still gives only $R@1 = 0.247$, because many unrelated candidates have comparable delays. Waveform embedding and sky-localization scores supply complementary information, and the catalog ranking score gives near-unity top-ten recall for both lens families. Figure 2 summarizes the single-score diagnostics and the catalog ranking score, and Fig. 3 shows the same separation in feature space.

The same validation-selected score was then evaluated over all unordered event pairs in the ET-3 test catalog (Fig. 4). This pair-level table contains $\binom{9000}{2} = 40,495,500$ pairs, of which 3,000 are true lensed doublets. The unordered pair score uses the larger of the two directed row-standardized scores, since the retrieval score is defined per query event. Because true negatives dominate this low base-rate problem (a positive rate of 7.41×10^{-5}), the pair-level ROC is saturated and is not used as the main diagnostic; precision–recall is the relevant summary. The pair-level AUPRC is 0.835. At the threshold closest to 50% pair recall, the catalog retains 1,500 true pairs and 201 false pairs, giving precision 0.882 at a false-positive rate of 4.96×10^{-6} .

We further thresholded the final unordered-pair score to quantify the follow-up burden directly (Fig. 4c). In the labeled ET-3 pair table, false-pair exceedance was computed only after removing the 3,000 true companion pairs. A target false-pair exceedance of 10^{-5} retained 2,619 true pairs and 405 false pairs. This gave pair recall 0.873 and precision 0.866. Thresholds chosen near 50% and 90% pair recall retained 201 and 505 false pairs, respectively. The corresponding precisions

were 0.882 and 0.842.

The same thresholding logic was then applied to a GWTC-3 real-background diagnostic with 200 synthetic injected true-pair rows. At target false-pair exceedance rates of 10^{-2} , 10^{-3} and 10^{-4} , the injected-pair recalls were 0.605, 0.235 and 0.035. The corresponding false-pair counts were 524, 53 and 6. For the real GWTC-3 and O4a catalogs, there are no confirmed lens labels. We therefore treat all real pairs as a null background. The top ten entries correspond to within-catalog tail fractions of $10/1953 = 0.00512$ for GWTC-3 and $10/3486 = 0.00287$ for O4a. These threshold exceedances define candidate shortlists for Bayesian follow-up, not detections or false-discovery rates.

Complete-pair recovery is limited by density and rarity

We next tested whether high-ranking edges can be called as complete lensed pairs, a harder task than companion retrieval because every accepted edge competes against order N^2 unrelated pairs. Even a small per-pair error rate can dominate the catalog. We report pair recovery using precision, thresholded false-pair exceedance and fixed budget shortlist behavior (Fig. 5; Methods).

The high recall reported above is obtained on a catalog far sparser and far richer in lensed events than an astrophysical 3G search. We therefore stress-tested ranking on observable-summary catalogs that reproduce the lensing physics of the main simulations. At the highest tested density, $10^5 \text{ events yr}^{-1}$, the time delay plus sky localization score gave top 10 recall 0.398 ± 0.110 (Fig. 5a). A calibrated surrogate waveform embedding improved this to 0.663 ± 0.062 . A weak combined score reached 0.475 ± 0.093 . The median companion rank was about 164 for time delay plus sky localization, compared with 4 for the surrogate waveform embedding.

The astrophysical strong-lensing fraction is expected to be of order 10^{-3} [35–37]. At this rarity, false associations become the decisive quantity, so precision and false-discovery measures directly track the catalog burden [39, 40]. In a top 50 fixed budget shortlist at lens fraction 10^{-3} , time delay plus sky localization recovered no true pairs. The surrogate waveform embedding alone gave precision 0.187 ± 0.110 and recall 0.173 ± 0.102 . The weak combined score increased these values to 0.300 ± 0.265 and 0.277 ± 0.249 . At lens fraction 10^{-4} , only the surrogate waveform embedding retained non-zero recovery, with precision 0.147 ± 0.061 and recall 0.137 ± 0.058 .

These sandbox tests show that waveform sequence information can mitigate the density limit of observable-only ranking. They also show that fixed budget recovery remains sensitive to the fusion rule at very low lens fractions. The large catalog waveform curves use calibrated surrogate embeddings rather than full strain encoders. They therefore support candidate generation design, not detection significance. Confirmation at realistic rarity still requires a discriminating follow-up stage

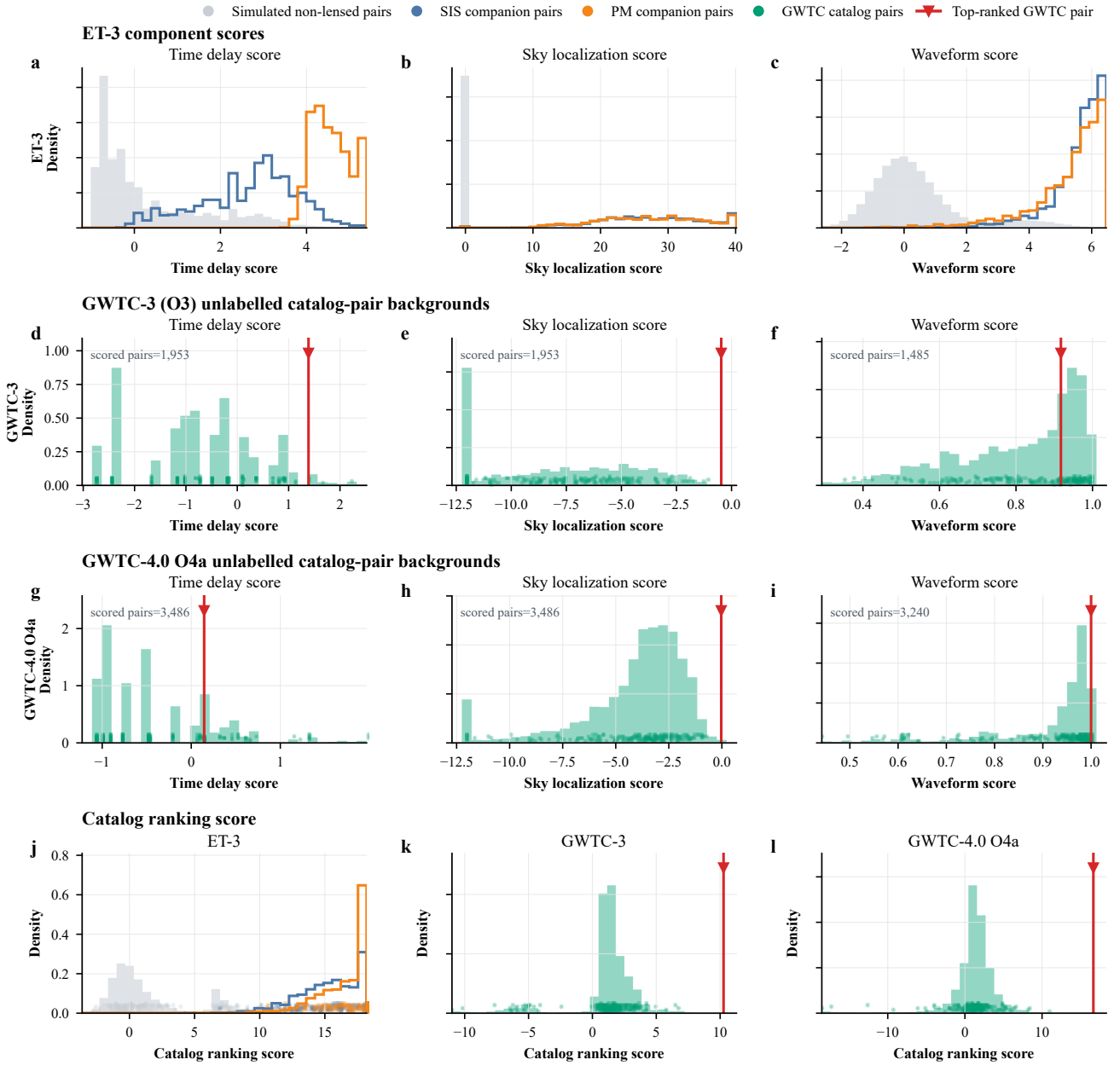


Figure 3. Pair-level score separation and real-catalog score backgrounds. (a–c) ET-3 time-delay, sky-localization and waveform embedding score distributions for true SIS companion pairs, true PM companion pairs and sampled non-lensed/background pairs. PM pairs separate most strongly in time, while SIS pairs rely more on sky-localization and waveform embedding scores. (d–f) GWTC-3 O3 time-delay, sky-localization and waveform embedding score backgrounds. (g–i) GWTC-4.0 O4a time-delay, sky-localization and waveform embedding score backgrounds. These GWTC panels show unlabelled catalog pairs rather than ranked lensed pairs. (j–l) Catalog ranking score distributions for ET-3, GWTC-3 and GWTC-4.0 O4a. The two GWTC catalogs are treated as empirical null backgrounds. Labels give scored event pairs rather than events; waveform embedding panels include only pairs with deployable waveform embeddings. Red markers denote the top-ranked GWTC pair in each catalog. The GWTC catalog pairs define follow-up context only and are not lensing detections.

such as full Bayesian posterior overlap or joint parameter estimation[22, 24, 25].

GWTC backgrounds test candidate refinement

We also tested the lens-candidate refinement stage on observed GWTC event backgrounds. GWTC-3 and the GWTC-4.0 O4a release are treated as two independent real-catalog deployments of the same work-

flow, each with a run-matched real-noise waveform encoder. The GWTC-3 deployment keeps the existing O3-noise encoder and uses 63 events with parameter-estimation support, giving 1,953 unordered pairs[4]. For O4a, we queried a later GWOSC O4 event API endpoint, which returned 140 public entries. This API count includes entries beyond the 128 new GWTC-4.0 O4a compact-binary candidates reported in the catalog paper. We therefore define the analysis sample by parameter-estimation support and real HEALPix

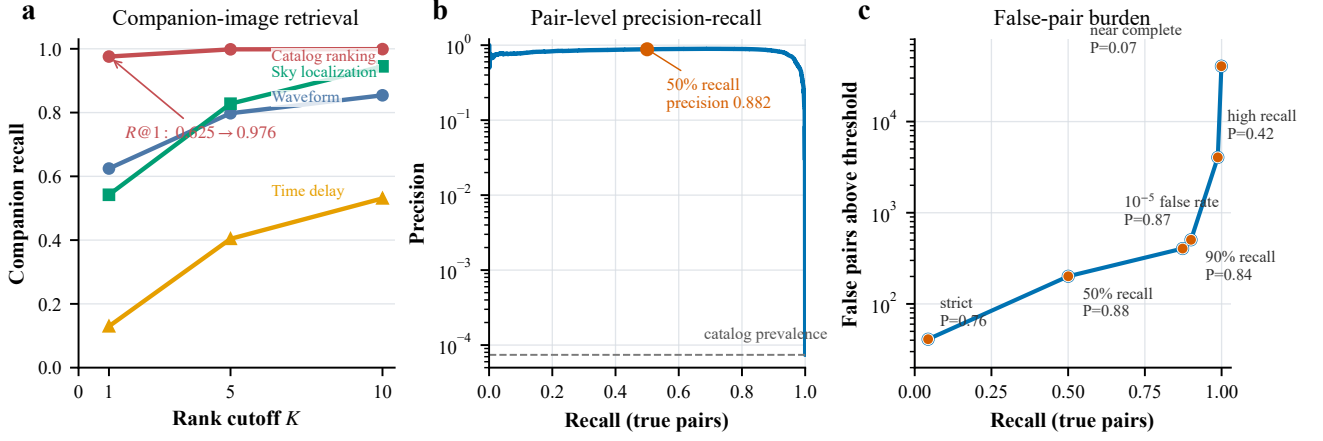
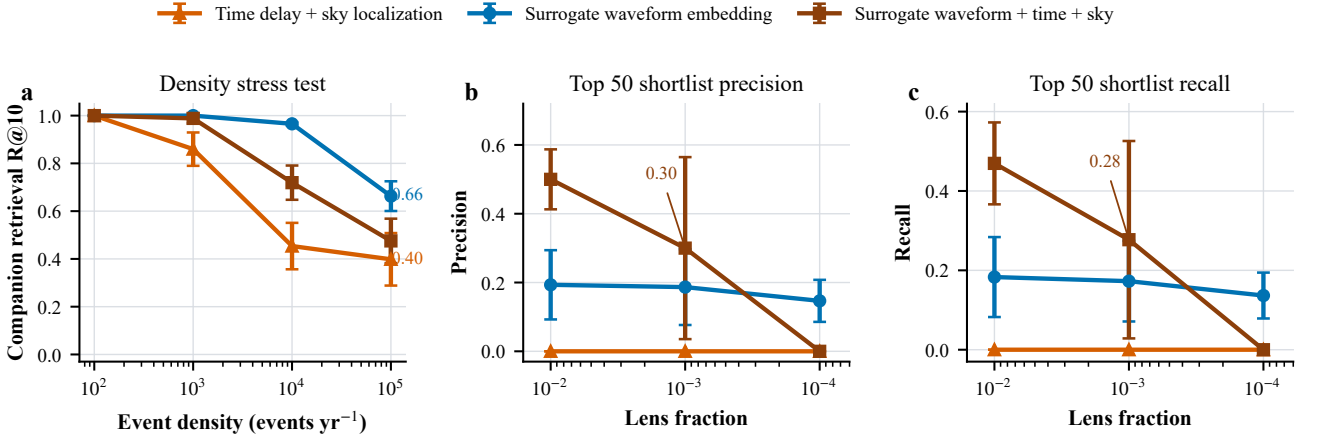


Figure 4. Retrieval gains must be read together with false-pair burden. (a) Companion-image recall as the rank cutoff K is relaxed for the waveform embedding, time-delay, sky-localization and catalog ranking scores. (b) Unordered pair-level precision–recall curve for all 40,495,500 ET-3 test-catalog event pairs. The orange marker denotes the threshold closest to 50% pair recall. (c) False-pair burden as a function of true-pair recall in the same labeled ET-3 pair table. Point labels give the precision at representative operating points.



Waveform curves use calibrated surrogate embeddings, not full strain encoders for large catalogs.

Figure 5. Calibrated waveform sequence information mitigates observable-only density limits. (a) Full catalog top 10 companion recall as the background event density increases at fixed lens physics. (b,c) Precision and recall for a top 50 shortlist at decreasing lens fraction. Points and error bars show the mean and seed-to-seed standard deviation over three seeds. The waveform curves use calibrated surrogate waveform embeddings matched to the ET-3 waveform embedding baseline, not full strain encoders for large catalogs. The combined curve gives the surrogate waveform score dominant weight and adds a weak time delay plus sky localization term.

sky maps, retaining 84 primary events and 3,486 un-ordered pairs[20, 41]. Public H1–L1 strain is sufficient to compute waveform embeddings for 55 GWTC-3 events and 81 O4a events, corresponding to 1,485 and 3,240 waveform-scored pairs.

Every recall number in these deployments is measured on synthetic lensed pairs injected into real off-source detector noise, not on the GWTC events themselves. The real catalogs contain no confirmed lenses and therefore provide no ground truth for recall. The GWTC-3 waveform embedding gate passes with macro $R@10 = 0.914$. Relative to the time-delay plus HEALPix sky-localization baseline, adding the waveform embedding score raises first-rank injection recall from 0.881 to 0.983, while both configu-

rations reach $R@10 = 1.000$. For the GWTC-4.0 O4a deployment, the O4-noise run-matched encoder is trained on 600 systems per family for 20 epochs. It passes the same gate with macro $R@10 = 0.858$ and minimum-family $R@10 = 0.844$. Validation selects a constrained three-score catalog ranking with non-zero waveform embedding, time-delay and HEALPix sky-localization weights. Relative to the time-delay plus sky-localization baseline, the catalog ranking score raises first-rank injection recall from 0.269 to 0.650 and top-ten recall from 0.658 to 0.983.

The lower O4a first-rank recall reflects the held-out validation set used for that deployment. In these injections, true-pair time separations overlap random-pair separations much more strongly than in the GWTC-

3 validation set. The time-delay plus sky-localization baseline is therefore less discriminative for O4a. The waveform embedding score still changes the practical retrieval regime: it moves the macro median rank from 5 to 1 and restores top-ten recall from 0.658 to 0.983. The selected O4a weights prioritize this top-ten follow-up recovery; larger waveform embedding weights gave higher R@1 but lower R@10.

The real GWTC pairs are then ranked as empirical null backgrounds. The highest-ranked GWTC-3 pair is GW191219_163120–GW191230_180458, and the highest-ranked O4a pair is GW230726_002940–GW230814_230901. Their waveform embedding, time-delay, sky-localization and catalog ranking score backgrounds are shown in Fig. 3, and the top 10 shortlists are shown in Fig. 6. Both top pairs have deployable waveform embedding scores. The O4a top pair is a BBH pair with waveform embeddings for both events and is not flagged as out of distribution. In the O4a deployment audit, all 3,240 BBH pairs with waveform embeddings have unique waveform embedding scores, with a standard deviation of 0.123. The historical GW170104–GW170814 candidate has a separation of 222.01 days and ranks 272nd of 1,953 GWTC-3 pairs under the catalog ranking score. These ranks define follow-up context only; they are ranked catalog coincidences, not lensing detection significances.

Discussion

We formulated strongly lensed gravitational-wave identification as a full-catalog retrieval problem. This framing is different from testing a pre-selected pair, because the search object is a catalog in which unrelated event pairs vastly outnumber true lensed systems. The neural encoder first gives each query event a waveform embedding score against its neighbours. The lens-candidate refinement stage then checks whether these neighbours also have lensing-like time-delay and sky-localization consistency. The resulting catalog ranking score is interpretable at the level of individual observables and can be used to rank candidate pairs for follow-up.

The ET-3 results show that the waveform encoder is essential but not sufficient. The waveform embedding score alone retrieves many companions, yet noisy full-catalog ranking still leaves enough near-neighbour confusion to limit first-rank recovery. Conversely, the time-delay plus sky-localization score without waveform embedding reaches only 71.5% first-rank recall, compared with 97.6% for the catalog ranking score. The auxiliary observables resolve different ambiguities. The time-delay score is most effective for the point-mass population, whose delay distribution is more distinctive in the simulated mass range. The sky-localization score contributes for both lens families, especially when time delays overlap the unrelated-event background. The validation-selected catalog ranking score therefore provides a practical compromise: it keeps the score transparent, allows weak components to be down-weighted, and performed more stably than the tabular learned

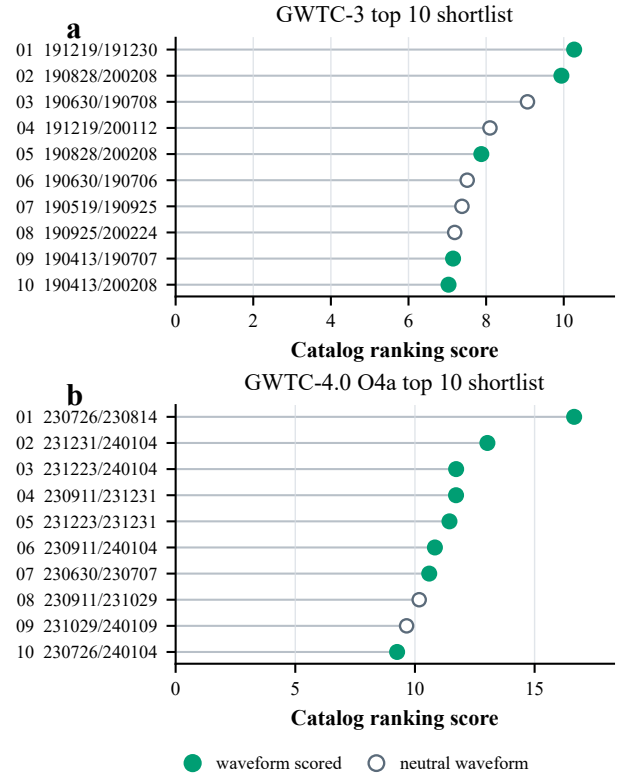


Figure 6. Real-catalog candidate shortlists on GWTC-3 and the GWTC-4.0 O4a release. (a,b) Top 10 real-catalog pairs under the catalog ranking score for GWTC-3 and GWTC-4.0 O4a. Filled markers denote pairs with deployable waveform embeddings, and open markers denote neutral waveform embedding contributions. The real catalogs are treated as unlabeled empirical backgrounds. These pairs are ranked catalog coincidences for follow-up and have not passed Bayesian lensing confirmation.

rankers tested here.

A central design choice is that the ranking stage uses only quantities that would be available in a search. True sky positions are used to generate simulated measurements but are not passed to the ranking stage. The sky-localization score is built from a network-SNR-dependent localization area and an observed sky center, which are then converted to pairwise localization scores. This prevents an oracle sky feature from defining the main result. It also sets a clear boundary on the present analysis. The current sky model is a compact A_{90} -based surrogate; fast localization posteriors or HEALPix sky maps are needed before the same logic can be used in a production search.

The catalog stress tests show why high retrieval recall should be interpreted with care. At low event density, the time delay and sky localization windows are sparse, so the true companion is often isolated among unrelated events. As the background density approaches third-generation rates, unrelated events increasingly occupy the same windows and the top 10 retrieval rate falls. The calibrated surrogate sandbox shows that waveform sequence information can mitigate this loss, but it does not remove the low base rate problem. At realistic lens rarity, pair-level false associations dominate the

accepted edge set even when the per-pair false-positive rate is small. The method can reduce the number of pairs requiring expensive confirmation, while final lens claims still need posterior-overlap, phase-consistency or joint-parameter-estimation analyses.

The GWTC tests support this refinement interpretation on observed catalog backgrounds. They also define the present boundary. GWTC-3 and the GWTC-4.0 O4a sample use separate run-matched encoders rather than a single shared embedding space, so we report the two catalogs separately. A merged cross-run ranking would require re-embedding all events with one encoder and recalibrating the score distributions across observing runs. The scale-aware preprocessing audit is important because an absolute normalization floor can collapse public strain inputs and make all waveform embedding scores identical. After correction, the waveform embedding score improves held-out real-noise injection validation in both deployments, and the O4a waveform embedding score audit no longer shows a collapsed score distribution. The lower O4a first-rank recall is a validation-domain limitation, while its top-ten recovery remains suitable for follow-up shortlisting. The real catalogs still provide only empirical null contexts. The historical GW170104–GW170814 pair is not promoted to the top of the present GWTC-3 ranking. Its long time separation is disfavoured by the time-delay score, which is a useful guard against background coincidences but could down-rank true lenses with unusually long delays.

Several extensions are needed before deployment. First, the ranked candidate list should be coupled to a fast confirmation stage and evaluated under a fixed follow-up budget. Second, the sky-localization score should be computed from rapid localization posteriors across realistic detector networks, with systematic checks of localization area and sky-error calibration. Third, the waveform encoder should be validated with larger real-noise injection campaigns and under network changes such as H1–L1, ET and ET–CE configurations. Finally, learned rankers need stricter negative construction, calibration and family-aware validation before they can replace the transparent catalog ranking score.

Taken together, the results identify where machine learning is useful in a lensing search. The encoder supplies scalable waveform neighbours, and the lens-candidate refinement stage makes the candidate set lens-aware. This division keeps the search computationally feasible while preserving the statistical boundary between candidate generation and lens confirmation.

Methods

Simulated catalogs

Compact-binary sources are generated with component masses uniform in $[10, 70] M_{\odot}$, isotropic sky positions, redshifts drawn uniformly in comoving volume out to $z = 2$, and geocenter arrival times uniform over an ap-

proximately ten-year window. We simulate two lens families. Singular-isothermal-sphere (SIS) lenses produce two opposed images. Their magnifications are set by source–lens alignment, and their Fermat delay is computed from the lens velocity dispersion ($100\text{--}300 \text{ km s}^{-1}$). Point-mass (PM) lenses, with lens masses sampled from $10^4\text{--}10^{10} M_{\odot}$, provide a complementary image geometry with a distinct time delay [5, 7, 10, 11]. PM systems are retained only when the interimage delay is at least 24 s and both image magnifications satisfy $|\mu| \geq 1$. These controlled SIS and PM parameter ranges were chosen with reference to the delay and magnification regimes represented in the GW-LMC catalog of Li, Liao and collaborators [42]. Each lensed system contributes two images. Isolated events form the singleton background. For each lens family we generate 10,000 systems, and we also generate 10,000 isolated events. Each family is split into 70% training, 15% validation and 15% testing. The mixed ET-3 validation and test catalogs each contain 1,500 SIS systems, 1,500 PM systems and 3,000 isolated events. This gives 9,000 events in total and 6,000 lensed query images.

Strains are projected onto the detector network and colored Gaussian noise at the target sensitivity is added [43–45]. The principal catalog uses the Einstein Telescope three-detector triangular configuration, whose three co-located interferometers we denote ET-3 [30, 31]. This is not a multi-site network. The median 90% localization area we adopt (~ 20 square degrees for ET-3) is specific to this configuration and source population, and should be read against the known dependence of gravitational-wave localization on detector network geometry and parameter-estimation method [46–49].

Before training the encoder, we inspected the simulated observables that later define the physical refinement stage. The PM and SIS populations occupy different regions of the interimage-delay distribution, so the time-delay score can add information beyond the waveform embedding score (Fig. 7). The detected ET samples also span broad signal-to-noise-ratio distributions for isolated, PM-lensed and SIS-lensed events (Fig. 8). These checks document the simulation and detector-response regime rather than a separate detection result.

Neural retrieval

Each event’s multi-channel strain is encoded by an InceptionTime convolutional network [50] into a fixed-length embedding, following the broader use of machine learning for gravitational-wave detection, characterization and data-quality problems [51–54]. The encoder is trained with a contrastive objective that pulls together embeddings of images from the same source and pushes apart unrelated events. In the mixed ET-3 run, each training epoch contains 14,000 lensed image pairs and 14,000 isolated self pairs. After training, all catalog embeddings are computed once. The waveform embedding score is the inner product of L2-normalized embeddings. Exhaustive evaluation is $O(N^2)$ and is used

Time delay between lensed images

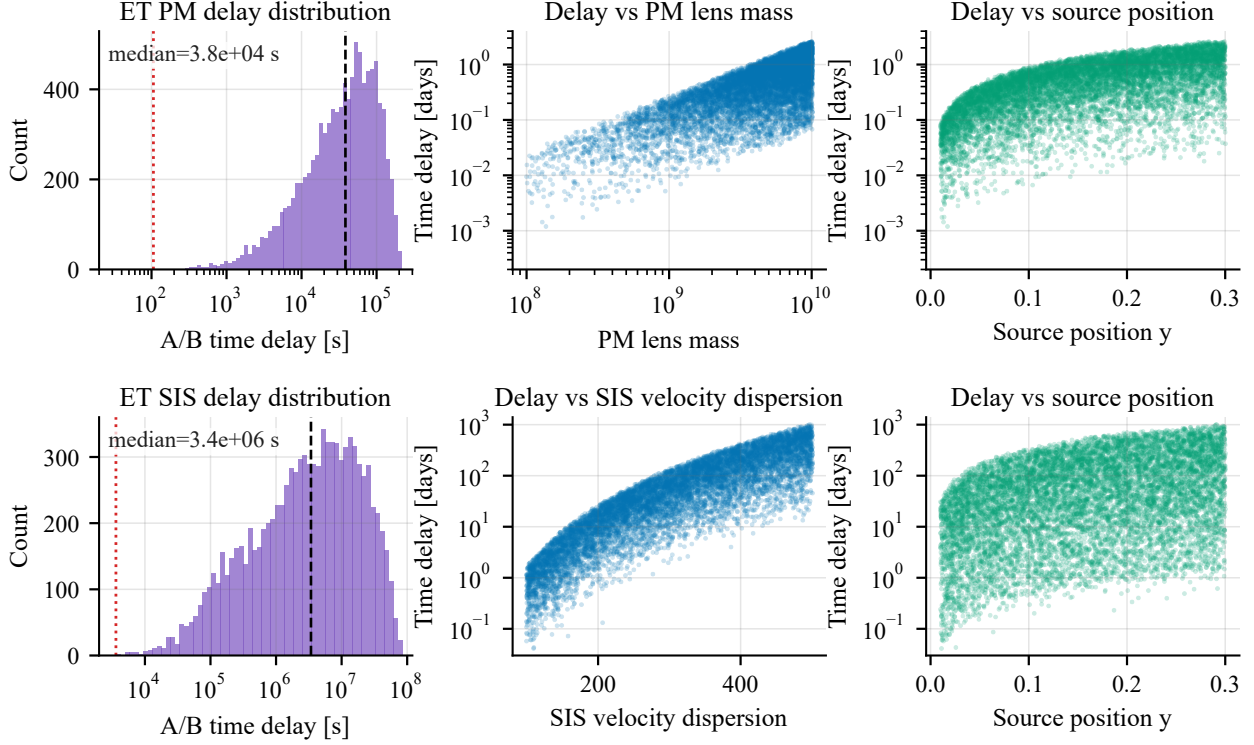


Figure 7. Time-delay structure of the simulated lens populations. Interimage-delay distributions and lens-parameter dependences for the PM and SIS systems used in the ET-3 catalog simulation. The separation between the two lens families motivates using a population time-delay score in the lens-candidate refinement stage.

Optimal SNR distributions after ET selection

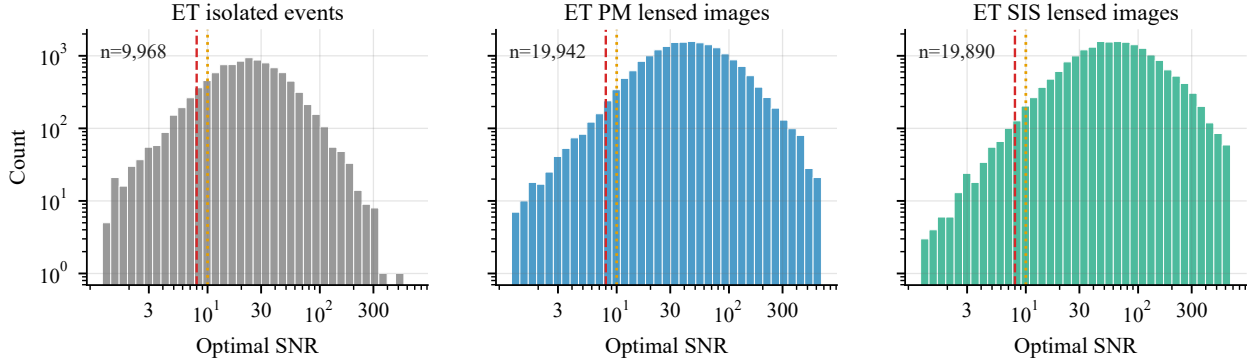


Figure 8. Signal-to-noise ratios in the simulated ET catalog. Optimal signal-to-noise-ratio distributions for isolated events and for PM- and SIS-lensed images after the detection selection.

here for the $N \leq 9,000$ catalogs where it is exact and tractable. Thus the main ET-3 ranking scores the dense non-self event-pair matrix. For larger catalogs, HNSW retrieval[55, 56] is used as a waveform-candidate generator before lens-candidate refinement. The retained-neighbour budget is chosen for a given catalog by comparing HNSW candidates with exact dense retrieval on tractable subsets and requiring stable dense-neighbour recovery and partner recall. After this prefilter, only $O(NK)$ candidate edges are scored, where K is the selected per-event follow-up budget. Self-matches are excluded; because embeddings are L2-normalized, the

inner product equals cosine similarity.

Lens-candidate refinement

In the dense ET-3 evaluation, the refinement stage scores every non-self event pair. In large-catalog variants, the same scoring can be applied to HNSW-preselected candidates. The refinement stage builds three pairwise scores computed from observable quantities. The time-delay score is a log likelihood ratio between a lensed-image delay distribution and a random catalog delay distribution. For the lensed distri-

bution we use the public GW-LMC mock catalog of Li, Liao and collaborators, which was built to provide signal-to-noise and interimage-delay priors for lensed gravitational-wave identification[42]. In the ET-3 analysis we use the ET BBH Any-Detected-SNR8 image table, retain systems with at least two detected images, and form all detected image pairs. This extraction gives 4,388 delays, with median 37.38 days and 90th percentile 284.70 days. The background delay distribution is estimated from random non-partner pairs in the held-out validation catalog. The binned log ratio is then fixed and applied to observed trigger-time differences at test time; injected delays are used only to generate the simulated catalog. The observed time is the true geocenter time plus jitter that scales with signal-to-noise ratio.

The sky-localization score is computed from an observed sky proxy rather than from the injected sky position or a learned sky prediction. For each simulated event, the injected sky coordinate is used only to generate an observed localization summary. Given the network signal-to-noise ratio ρ_i , we assign a 90% localization area $A_{90,i} = A_{90,\text{ref}}(\rho_{\text{ref}}/\rho_i)^2\eta_i$, where η_i is a lognormal scatter term. We then apply detector-dependent clipping. This area is converted to an effective Gaussian width σ_i by matching the 90% containment area of a two-dimensional tangent-plane Gaussian. The observed sky center is drawn around the injected direction with this width.

Pairwise sky-localization consistency is computed only from the observed centers and widths. For events i and j , we calculate the observed angular separation θ_{ij} and the combined width $\sigma_{ij} = (\sigma_i^2 + \sigma_j^2)^{1/2}$. The normalized separation $d_{ij} = \theta_{ij}/\sigma_{ij}$ defines the windowed sky-localization score used in the main ET-3 channel analysis. We also test a Gaussian log-overlap score, $-\log(2\pi\sigma_{ij}^2) - \theta_{ij}^2/(2\sigma_{ij}^2)$, in supporting analyses. True sky coordinates are never passed to the ranking stage. We also tested an amplitude channel based on signal-to-noise-ratio consistency, including a two-dimensional delay-SNR-ratio likelihood ratio extracted from the same GW-LMC table. This amplitude/SNR channel is a supporting analysis and is not used in the main reported score unless stated explicitly.

The catalog ranking score is a weighted sum of row-standardized waveform embedding, time-delay and sky-localization scores, $S_{ij} = \sum_m \lambda_m z(\text{score}_m)_{ij}$. We select the weights on a held-out validation catalog over a grid that includes zero, allowing a component to be switched off. The weights are then fixed for testing. The waveform embedding weight is also tuned and is driven to zero on the noisy LIGO catalog. We benchmark this catalog ranking score against logistic regression, histogram gradient boosting, a tabular multilayer perceptron and a gradient-boosted ranker on the same features[57, 58].

Thresholded false-pair diagnostics use the same unordered-pair catalog ranking score. In labeled ET-3 catalogs, we remove the 3,000 true companion pairs before estimating the false-pair exceedance rate. Recall

and precision are then computed by applying the same threshold to the true and false pair sets. In the GWTC injection diagnostic, the true rows are the synthetic injected lensed pairs. The false rows are real-background pairs. In the real GWTC catalogs, no confirmed lens labels are available. We therefore treat all real pairs as a null population. We report only within-catalog tail fractions and shortlist sizes, not precision, false-discovery rate or detection significance.

Observable-level stress-test catalogs

The density and rarity stress tests require catalogs far larger and rarer than the strain-level simulations, for which generating full waveforms is impractical. We use an observable-summary generator that reproduces the source priors and SIS/PM lensing equations of the main simulations (magnifications, Fermat delays) but emits only the observable summaries used by the ranking stage. Network signal-to-noise is modeled with the correct leading-order scaling. It is calibrated so the median localization area matches the strain pipeline. Each image's observed sky position is drawn independently around its true direction, with a width set by its localization area. The only quantity not reproduced exactly is the matched-filter signal-to-noise ratio. The stress tests depend mainly on the relative time, sky and mass structure, which are reproduced at observable level, so their conclusions are insensitive to this approximation.

For the sequence information sandbox, we use the same observable-summary generator but add a calibrated surrogate waveform embedding. Each source is assigned a latent waveform center. Independent encoder noise is tuned so that full-catalog waveform embedding retrieval matches the measured ET-3 baseline, $R@1 = 0.625$ and $R@10 = 0.854$. This surrogate is used only to test large catalog shortlisting behavior. It is not a full strain waveform encoder for 10^5 -event catalogs.

For the density test, each of three random seeds requests 60 lensed doublets and scales the unlensed background at fixed lens physics. The requested background densities are 10^2 , 10^3 , 10^4 and 10^5 events yr^{-1} over the same observation window. After the signal-to-noise cut, the mean detected catalog sizes over the three seeds are 590, 4,998, 49,080 and 489,474 events. Companion-image recall is computed over detected lensed images that retain a detected partner.

For the rarity test, we use the same lensed population and size the background to nominal lens fractions of 10^{-2} , 10^{-3} and 10^{-4} . After detection cuts, each seed contains on average 53.7 complete true pairs. The mean detected catalog sizes are 2,972, 28,802 and 286,857 events for the three lens-fraction labels. We compare time delay plus sky localization, the calibrated surrogate waveform embedding alone and a weak combined score, $z(\text{waveform}) + 0.01 z(\text{time} + \text{sky})$. We report fixed global shortlist budgets of 50, 100 and 200 pairs. The false-pair tail is estimated by Monte Carlo sampling 350,000 false pairs per seed and scaling to the full unordered-pair count.

Parameter-scatter sandbox check

For the parameter-scatter check we compare three retrieval scores after adding independent per-event scatter to posterior-summary parameters. The parameter-summary baseline ranks events by nearest-neighbor distance in chirp mass, mass ratio and luminosity distance. The waveform embedding score and the time-delay plus sky-localization score do not use these intrinsic parameter summaries, so their scores are held fixed while the scatter is varied.

GWTC event analysis

For the GWTC analyses, the primary real catalog is defined by parameter-estimation support and an available HEALPix sky map. Waveform availability is treated as a separate audit axis. The GWTC-3 deployment uses 63 compact-binary events, giving 1,953 unordered pairs. The O4a deployment corresponds to the GWTC-4.0 data release, which reports 128 new O4a compact-binary candidates. Our local GWOSC query used a later O4 event API endpoint and returned 140 public entries before analysis cuts. Applying the same parameter-estimation and sky-map requirements retains 84 primary events, giving 3,486 unordered pairs. Events outside the binary-black-hole training prior are retained in the catalog but flagged as out of distribution; pairs containing such events are not used to support waveform embedding claims.

For each real catalog, the ET-3 encoder is not reused directly. We use public H1-L1 strain and train run-matched InceptionTime encoders on synthetic SIS and PM lensed pairs injected into real off-source detector noise. The GWTC-3 result keeps the existing O3-noise encoder. The O4a encoder is trained on 600 systems per family for 20 epochs using O4a off-source H1-L1 strain injections. A separate O3-to-O4 transfer test is recorded only as an ablation. The strain preprocessing uses per-channel scale-aware standardization, since a fixed absolute normalization floor can suppress public strain amplitudes and collapse the embeddings. A Gate-1 test on held-out real-noise injections is required before using the waveform embedding score. The pass criterion is macro $R@10 > 0.6$ and minimum-family $R@10 > 0.5$.

The real-catalog ranking uses three scores: waveform embedding, time-delay and HEALPix sky-localization scores. Each HEALPix map is converted to RING ordering, resampled to a common $N_{\text{side}} = 512$, clipped to finite non-negative values and normalized so that $\sum_p P_{ip} = 1$. For events i and j , the sky-posterior overlap underlying the sky-localization score is

$$O_{ij}^{\text{sky}} = \frac{\sum_p P_{ip} P_{jp}}{\left(\sum_p P_{ip}^2\right)^{1/2} \left(\sum_p P_{jp}^2\right)^{1/2}}, \quad (1)$$

where p indexes equal-area HEALPix pixels. This expression is the cosine overlap of the two posterior probability vectors; the pixel-area factors cancel after normal-

ization. The HEALPix sky-localization score is $s_{ij}^{\text{sky}} = \log[\max(O_{ij}^{\text{sky}}, 10^{-300})]$. Signal-to-noise-ratio and amplitude quantities are retained only as audit columns and are not used in the catalog ranking score. We select ranking weights separately for each deployment on held-out real-noise injection validation catalogs over the grid $\{0, 0.25, 0.5, 1, 2, 4\}$. The reported real-catalog deployments use constrained positive three-score catalog ranking, so waveform embedding, time-delay and sky-localization terms all contribute to the catalog ranking score. The selected GWTC-3 weights are 0.25, 0.25 and 4.0 for waveform embedding, time-delay and sky-localization scores. The selected O4a weights are 1.0, 0.25 and 4.0. Exact grid-selected values are recorded in the released run metadata. Each catalog ranking score is fixed before scoring the corresponding real catalog: 1,953 GWTC-3 pairs and 3,486 O4a pairs. Public strain is sufficient to compute waveform embeddings for 55 of the 63 GWTC-3 events and 81 of the 84 O4a events, corresponding to 1,485 and 3,240 waveform-scored pairs. The remaining 468 GWTC-3 pairs and 246 O4a pairs have at least one event without a deployable embedding. For these pairs, the row-standardized waveform embedding term is set to zero, equivalent to assigning the row mean in the waveform embedding score. Their ranking is therefore determined by the available time-delay and HEALPix sky-localization terms, while the full catalog remains defined. The shortlist records whether each pair has a deployable waveform embedding score.

Data availability

The gravitational-wave event data analysed here are publicly available from the Gravitational Wave Open Science Center (<https://gwosc.org>) and the associated parameter-estimation data releases on Zenodo. The simulated catalogs and derived observable tables generated in this study are available in the project repository (<https://github.com/lanhung/gw-catalog>).

Code availability

All analysis code, including neural retrieval, lens-candidate refinement, sparse retrieval and stress-test pipelines, is openly available at <https://github.com/lanhung/gw-catalog>. Some internal script filenames retain historical labels such as “gwtc5” or “gwtc4p1”, but the manuscript text refers to the corresponding GWTC-4.0 O4a sample consistently.

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Author contributions

F.Z. and X.F. designed the study. F.Z. and Q.Z. developed the retrieval and catalog ranking pipeline and performed the experiments. Q.Y., J.H. and Y.Y. contributed to the simulations, analysis and validation. F.Z. and X.F. wrote the manuscript. All authors reviewed and approved the manuscript.

Competing interests

The authors declare no competing interests.